

Nick Eisenbergs Comprehensive Presentation

Fall 2020

Gaussian Processes

A Gaussian Processes, $G := \{G_t : t \in T\}$ is a stochastic process such that for any finite set of $t_1, \dots, t_n \in T$, the random vector $(G(t_1), \dots, G(t_n))$ follows a n -dimensional Gaussian distribution. The processes G is uniquely determined by

1. $\mu(t) = \mathbb{E}(G(t))$
2. $C(s, t) = \mathbb{E}(G(t)G(s))$

Theorem (Important)

The space of all symmetric and non-negative definite functions, $f(s, t)$, on $T \times T$ matches the space of all covariance functions of all Gaussian processes on T . Thus, given a non-negative definite function on $T \times T$ then there exists a Gaussian Processes that possesses this covariance function.

Examples of Gaussian Processes

1. **Brownian Motion:** Consider the following mean and covariance functions on $T = [0, \infty)$.

- ▶ $\mu(t) = 0$
- ▶ $C(s, t) = \min(s, t)$

It can be shown that C is non-negative definite and the resulting Gaussian Processes is referred to as Brownian Motion,
 $B = \{B(t) : t \geq 0\}$.

2. **White Noise:** Consider the following mean and covariance functions on $T = \mathcal{B}(\mathbb{R}^N)$, the Borel sets of $\mathbb{R}^N$.

- ▶ $\mu(A) = 0$
- ▶ $C(A, B) = \lambda^N(A \cap B)$, $\lambda^N = N$ -dimensional Lebesgue measure.

Again it can be shown that C is non-negative definite and thus the resulting Gaussian processes on T is referred to as White Noise,
 $\dot{W} = \{\dot{W}(A) : A \in \mathcal{B}(\mathbb{R}^N)\}$.

Brownian Motion in terms of white noise

- ▶ Consider a white noise, $\dot{W} = \{\dot{W}(A) : A \in \mathcal{B}(\mathbb{R})\}$
- ▶ Define $B(t) = \dot{W}([0, t])$
- ▶ Since $\dot{W}$ is a white noise we have
$$\mathbb{E}(B_s B_t) = \mathbb{E}(\dot{W}[0, s] \dot{W}[0, t]) = \lambda([0, s] \cap [0, t]) = \min\{s, t\}$$
- ▶ This shows that B is a Brownian Motion

White Noise as an Ito Integral

We define a set function on $\mathcal{B}(\mathbb{R})$

$$\dot{W}(A) = \int_0^\infty 1_A(t) dB(t, \omega)$$

For any borel set, A , the deterministic indicator function $1_A \in L^2(\mathcal{P})$ and so the Ito Integral exists.

From the theorem in the previous slide we see that

- ▶ $\mathbb{E}\dot{W}(A) = 0$
- ▶ $\mathbb{E}(\dot{W}(A)\dot{W}(B)) = \int_0^\infty 1_A(t)1_B(t)dt = \lambda(A \cap B)$

Thus, $\dot{W}$ is a white noise on $\mathcal{B}(\mathbb{R})$

Ito Integral

We will briefly explain the meaning of $\int_0^\infty u(t, \omega) dB(t, \omega)$, the Ito Integral of u , where B is a Brownian Motion.

Definition (Progressively Measurable)

A process, $u(t, \omega)$, is progressively measurable if u restricted to $\Omega \times [0, t]$ is $\mathcal{F}_t \times \mathcal{B}([0, t])$ measurable where $\mathcal{F}_t = \sigma\{B(s) : 0 \leq s \leq t\}$.

Definition (Progressively Measurable Hilbert space, $L^2(\mathcal{P})$)

We define $L^2(\mathcal{P}) = L^2(\Omega \times [0, \infty), \mathcal{P}, \mathbb{P} \times \lambda)$ where $\mathcal{P}$ is the sigma-algebra generated by all the sets $A \subset \Omega \times [0, \infty)$ such that $1_A(t, \omega)$ is progressively measurable. The norm of this Hilbert Space is $\|u\|_{\mathcal{P}}^2 = \mathbb{E} \left(\int_0^\infty u(t, \omega)^2 dt \right)$

Definition (Simple Processes and their Ito Integral)

A process of the form $u(t, \omega) = \sum_{k=1}^n \phi_k(\omega) 1_{(t_k, t_{k+1}]}(t)$ where ϕ_k is $\mathcal{F}_k$ measurable is called a simple process. Its Ito integral is defined to be $\int_0^\infty u(t, \omega) dB(t, \omega) = \sum_{k=1}^n \phi_k(\omega) (B(t_{k+1}, \omega) - B(t_k, \omega))$

Ito Integral

Theorem

The simple processes are dense in $L^2(\mathcal{P})$. Thus for any processes, $u(t, \omega)$, there exists a sequence of simple processes, $u_n(t, \omega)$ such that $\mathbb{E} \left(\int_0^\infty (u(t, \omega) - u_n(t, \omega))^2 dt \right) \rightarrow 0$.

Definition (Ito Integral of a $L^2(\mathcal{P})$ processes)

For any processes $u(t, \omega) \in L^2(\mathcal{P})$ we define its Ito Integral to be

$$\int_0^\infty u(t, \omega) dB(t, \omega) := \lim_{n \rightarrow \infty} \int_0^\infty u_n(t, \omega) dB(t, \omega)$$

where the limit is with respect to the $L^2(\mathcal{P})$ norm and the u_n 's are from the Theorem.

Theorem

1. $\mathbb{E} \int_0^\infty u(t, \omega) dB(t, \omega) = 0$
2. $\mathbb{E} \left(\int_0^\infty u(t, \omega) dB(t, \omega) \cdot \int_0^\infty v(t, \omega) dB(t, \omega) \right) = \mathbb{E} \left(\int_0^\infty u(t, \omega) v(t, \omega) dt \right)$

Walsh Integral

$$\int_0^t \int_{\mathbb{R}^d} f(s, y) W(ds, dy)$$

- ▶ Generalization of the Ito Integral
- ▶ Defined for two parameter worthy martingale measure integrators.
 - ▶ The white noise processes $W_t(A) = \dot{W}([0, t] \times A)$ is "worthy" and is used in the 1- d SHE and SWE.
 - ▶ A "colored" Gaussian noise will be used in higher dimensions
- ▶ Used to define the solution to the SHE and SWE for two parameter noise forcing terms.

Walsh Integral Construction

- ▶ Simple functions: Let $\mathcal{L}$ be class of linear combinations of processes of the form $f(t, x, \omega) = X(\omega)1_{(a,b]}(t)1_A(x)$ where $X \in \mathcal{F}_a$ and let $\mathcal{P} = \sigma(\mathcal{L})$
- ▶ The integral for a simple processes is

$$(f \cdot M)_t(B) = X(\omega) [M_{b \wedge t}(A \cap B) - M_{a \wedge t}(A \cap B)]$$

Theorem

For $\mathcal{P}$ measurable processes, $f(t, x, \omega)$ that satisfy

$$\int_0^T \iint_{\mathbb{R}^{2n}} |f(x, t)f(y, t)|K(dx, dy, dt) < \infty$$

there exists a sequence of simple processes whose integrals are Cauchy in $L^2(\mathcal{P})$. The limit is the walsh integral of f .

Colored Noise

- ▶ For non-negative and non-negative definite measure Γ define on $\mathcal{S}(\mathbb{R}^d)$

$$\begin{aligned} C(\varphi, \psi) &= \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\varphi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \varphi(s, x) f(x - y) \psi(s, y) \end{aligned}$$

- ▶ C is non-negative definite and the resulting 0-mean Gaussian processes is referred to as a colored Gaussian noise, $F = \{F(\varphi) : \varphi \in \mathcal{S}(\mathbb{R}^d)\}$.
- ▶ Bochner's Theorem implies the existence of non-negative measure μ (spectral density) such that

$$\int \Gamma(dx) \varphi(x) = \int \mu(dx) \mathcal{F}\varphi(x)$$

Walsh integral with Colored Noise Integrator

- ▶ Take $\varphi_n \in \mathcal{S}(\mathbb{R}^d)$ such that $\varphi_n \rightarrow 1_A$
- ▶ Can be shown $\mathbb{E}[(F(\varphi_n) - F(\varphi_m))^2] \rightarrow 0$. Call the limit $F(A)$.
- ▶ $M_t(A) := F([0, t] \times A)$ is worthy and can be used as an integrator.
- ▶ Through the construction we can show that

$$F(\varphi) = \int_0^\infty \int_{\mathbb{R}^d} \varphi(s, x) M(dx, ds)$$

- ▶ The Walsh integral for integrator M can be defined for processes satisfying

$$\int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \varphi(s, x) f(x - y) \varphi(s, y) < \infty$$

Stochastic Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = \sigma(u(t, x))\dot{M}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The rigorous meaning of the SHE is

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x-y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y)\sigma(u(s, y))M(ds, dy)$$

- Informally the $\dot{M}$ is acting like the density (does not exist) of M in the sense that

$$\int_0^t \int_{\mathbb{R}^d} \varphi(s, y)M(ds, dy) = \int_0^t \int_{\mathbb{R}^d} \varphi(s, y)\dot{M}(s, y)dsdy$$

Definition of Invariant Measures

Consider an arbitrary measure space $(E, \mathcal{E})$ and denote $\mathcal{M}_1(E)$ to be the space of all probability measure on $(E, \mathcal{E})$.

Definition (Markovian Transition Function)

We say that $P_t(x, \Gamma)$, $t > 0$, $x \in E$, $\Gamma \in \mathcal{E}$, is a Markovian Transition Function is:

- (i) $P_t(x, \cdot)$ is a probability measure on $(E, \mathcal{E})$ for $t \geq 0$, $x \in E$
- (ii) $P_t(\cdot, \Gamma)$ is a $\mathcal{E}$ -measurable function for $t \geq 0$ and $\Gamma \in \mathcal{E}$
- (iii) $P_{t+s}(x, \Gamma) = \int_E P_t(x, dy) P_s(y, \Gamma)$
- (iv) $P_0(x, \Gamma) = 1_\Gamma(x)$ where $1_\Gamma(x)$ is the indicator function.

Invariant Measures viewed as an average

Definition (Invariant Measure)

A measure $\eta(\cdot) \in \mathcal{M}_1(E)$ is invariant for transition functions $\{P_t(x, \Gamma)\}_{t \geq 0}$ if for any $\Gamma \in \mathcal{E}$ we have that

$$\eta(\Gamma) = \int_E P_t(x, \Gamma) \eta(dx) \quad \text{for any } t \geq 0$$

Recall from Calculus that the average of a function $f : [a, b] \rightarrow \mathbb{R}$ is defined as

$$\text{AVG}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$

Thus, $\eta(\Gamma)$ is the average of the function $x \mapsto P_t(x, \Gamma)$ over $(E, \mathcal{E}, \eta)$.

Main Result

The laws of the solution the the SHE defines Markovian transition functions

$$P_t(\varphi, \Gamma) = \mathcal{L}(u^\varphi(t, \cdot))(\Gamma) = P[\omega \in \Omega : u^\varphi(t, \cdot) \in \Gamma] \quad \Gamma \in \mathcal{B}(L_\rho^2)$$

where $u^\varphi(t, x)$ is the solution to the SHE starting at $\varphi \in L_\rho^2$. We have the following main results

Theorem

Under certain conditions of the spectral density $\hat{f}$ and on the admissible weight $\rho(x)$, there exists a measure η that is invariant for the solution to the stochastic heat equation for any L_ρ^2 initial condition. In other words, for $P_t(\varphi, \Gamma)$ we have that, $\eta(\Gamma) = \int_{L_\rho^2(\mathbb{R}^d)} P_t(x, \Gamma) \eta(dx)$ for $t \geq T > 0$ for any $T > 0$.

Theorem

Under the same assumptions as the theorem above, there exists a measure η that is invariant for the solution to the stochastic heat equation with the $\delta_0(x)$ initial condition.

Importance of the Result/Comparison to Previous Results

- ▶ Translation of the paper [12] from Daprato theory to Walsh Theory.
- ▶ Existence of Invariant Measures for a the SHE with a broader range of initial conditions, in particular measure valued initial conditions such as δ_0 .
 - ▶ [12] has constant 1 in initial condition, Lipschitz diffusion coefficient and 0 drift term
 - ▶ [9] allowed L^2_ρ initial conditions with bounded and Lipschitz coefficients.
 - ▶ [1] Allows unbounded coefficients. The diffusion coefficient is Lipschitz and the drift has a polynomial type growth. The weight is given to be $\rho(x) = (1 + x^2)^{-e/2}$.

Current Project: Fractional SPDE

$$\begin{cases} (\partial^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2) \end{cases} \quad (0.1)$$

- ▶ $a \in (0, 2]$, $b \in (0, 2)$, $\nu > 0$, $r > 0$ and $\theta > 0$
- ▶ $(-\Delta)^{a/2}$ refers to the fractional Laplacian [8]
- ▶ ∂^b refers to the Caputo derivative [11]
- ▶ I_t^r is the Riemann-Liouville fractional integral operator.
- ▶ The time independent noise $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ arises from the covariance function

$$\mathbb{E}(W(\varphi)W(\psi)) = \langle \varphi, \psi \rangle_{\mathcal{H}} = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

Fractional Laplacian:

- **Fractional Laplacian:** One way to define the fractional Laplacian is through the Fourier Transform. It can be shown that

$$\mathcal{F}(-\Delta f)(\xi) = K|\xi|^2 \mathcal{F}f(\xi)$$

where K depends on the definition of the Fourier transform. So for $\psi \in \mathcal{S}(\mathbb{R}^d)$ we define

$$\mathcal{F}(\psi(-\Delta)f)(\xi) = \psi(K|\xi|^2) \mathcal{F}f(\xi)$$

and in particular we can show that if we want to solve

$$(-\Delta^{\alpha/2})u = f \implies u = (-\Delta^{-\alpha/2})f$$

then u is given by $f * R_\alpha$ where R_α is the Reisz Kernel.

$$R_\alpha(x) = \frac{\Gamma((1/2)(d - \alpha))}{\Gamma((1/2)\alpha) 2^\alpha \pi^{d/2} |x|^{d-\alpha}}$$

Fundamental solution to the deterministic FPDE

$$\begin{cases} (\partial^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [f(t, x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2) \end{cases} \quad (0.2)$$

$$u(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) + (Y \star f)(t, x) & \beta \in (0, 1] \\ (\frac{\partial Z}{\partial t}(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) + (Y \star f)(t, x) & \beta \in (1, 2) \end{cases}$$

- ▶ $Z(t, x) = \pi^{-d/2} t^{[\beta]-1} |x|^{-d} H_{2,3}^{2,1} \left(\frac{|x|^\alpha}{2^{\alpha-1} \nu t^\beta} \middle| \begin{matrix} (1, 1) & ([\beta], \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right)$
- ▶ $Y(t, x) = {}_t D_+^{[\beta]-\beta-\gamma} Z(t, x)$

Reason for constant 1 initial conditions

It can be shown that for the above initial conditions that

- ▶ $(Z(t, \cdot) * 1)(x) = 1$ when $\beta \in (0, 1]$
- ▶ $(\frac{\partial Z}{\partial t}(t, \cdot) * 1)(x) = 1$ when $\beta \in (1, 2)$

In addition, it can be shown that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(c^{b/a} \xi).$$

where as for the Wave equation we have

$$\mathcal{F}G(ct, \cdot)(\xi) = c \mathcal{F}G(t, \cdot)(c\xi)$$

and because of these similar scaling properties, we can almost treat Y and G as the same.

FSPDE Integral Equation

- ▶ The time independence leads us to rigourously defining the FSPDE through the following Skorohod integral equation

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} Y(t-s, x-y) u(s, y) W(dy) \right) ds$$

- ▶ Through Piccard iteration we show that

$$u(t, x) = 1 + \sum_{n \geq 1} \theta^{n/2} I_n(f_n(\cdot, x, t))$$

where

$$f_n(x_1, \dots, x_n, x, t) = \int_{[0, t]^n} G(t - t_n, x - x_n) \cdots G(t_2 - t_1, x_2 - x_1) dt$$

Moment Asymptotics for SWE

For the SWE with time independent noise the following two have been proven in [2]

- ▶ Existence and uniqueness of the solution.
 - ▶ Proven by showing $\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$
- ▶ Let $t_p = (p - 1)^{\frac{1}{4-\alpha}} t$. The following moment asymptotic has been proven for the SWE for dimensions $0 < \alpha < d \leq 3$

$$\lim_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p = \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2}\right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha}\right)^{\frac{4-\alpha}{3-\alpha}}$$

FSPDE

- ▶ Existence and uniqueness of the solution
- ▶ Similar asymptotics

References I

 Assing, Sigurd and Manthey, Ralf.

Invariant measures for stochastic heat equations with unbounded coefficients

Stoc. Proc. and their App. 2002.

 Balan, R., Chen, Le and Chen, Xia.

Exact asymptotics of the stochastic wave equation with time-independent noise.

 Chen, Le and Jingyu Huang.

Comparison principle for stochastic heat equation on $\mathbb{R}^d$.

Ann. Probab. 2019, to appear.

References II



Chen, Le and Kunwoo Kim.

Nonlinear stochastic heat equation driven by spatially colored noise: moments and intermittency.

Acta Math. Sci. Ser. B, 2019, to appear.



Dalang, Robert, Davar Khoshnevisan, Carl Mueller, David Nualart, and Yimin Xiao

A minicourse on stochastic partial differential equations.



Da Prato, Giuseppe and Jerzy Zabczyk.

Stochastic equations in infinite dimensions. Encyclopedia of Mathematics and its Applications, 44. Cambridge University Press, Cambridge, 1992. xviii+454 pp.

References III

-  Da Prato, Giuseppe and Jerzy Zabczyk.
Ergodicity for infinite-dimensional systems.
London Mathematical Society Lecture Note Series, 229. Cambridge University Press, Cambridge, 1996.
-  Folland, Gerald
Introduction to Partial Differential Equations.
-  Misiats, Oleksandr, Oleksandr Stanzhytskyi, and Nung-Kwan Yip.
Existence and uniqueness of invariant measures for stochastic reaction-diffusion equations in unbounded domains.
J. Theoret. Probab. 29 (2016), no. 3, 996–1026.
-  Nualart, D and Nualart, E.
Introduction to Malliavin Calculus
Cambridge. Cambridge University Press (2018)

References IV



Podlubny, L.

Fractional Differential Equations

Mathematics in Science and Engineering



Tessitore, Gianmario and Jerzy Zabczyk.

Invariant measures for stochastic heat equations.

Probab. Math. Statist. 18 (1998), no. 2, Acta Univ. Wratislav. No. 2111, 271–287.



Walsh, John B.

An Introduction to Stochastic Partial Differential Equations.

In: *École d'été de probabilités de Saint-Flour, XIV—1984*, 265–439.
Lecture Notes in Math. 1180, Springer, Berlin, 1986.